Final Term Assignment

Marks: 20 | Deadline: **20/05/2026 11:59 PM**

Topic: Supermarket Sales Analysis System

Use a real supermarket sales dataset to go through the entire data pipeline — clean it, explore it, find patterns, group customers, and predict behavior.

Step 1. Preprocessing

Download the free *Supermarket Sales* dataset from Kaggle. Clean it: handle any missing values, fix data types, normalize the numeric columns (unit price, quantity, rating).

Step 2. Data Overview

Identify which features are Nominal (product category, gender), Ordinal (rating level), and Ratio (total sales). Split data into 80% train / 20% test.

Step 3. Association Rules

Apply the Apriori algorithm to find which product categories are frequently bought together. Example finding: *"Customers who buy Health & Beauty also tend to buy Food & Beverages."*

Step 4. Clustering

Use K-Means (k=3) to group customers by their total spending and visit frequency. Label the groups: Low Value, Mid Value, High Value customers.

Step 5. Classification

Build a simple classifier (Decision Tree) to predict whether a customer will rate their experience above 7 (Good) or not, based on branch, product line, and payment method.

Dataset Description

Supermarket Sales — freely available on Kaggle (search: *"supermarket sales dataset"*). It has ~1,000 rows and 17 columns — small enough to work with in Excel or Python.

Submission:

1. Please submit a Python notebook on all the above tasks with as much comments and details as possible. Please also specify 2-sentence interpretation of each result in the last block of the notebook.